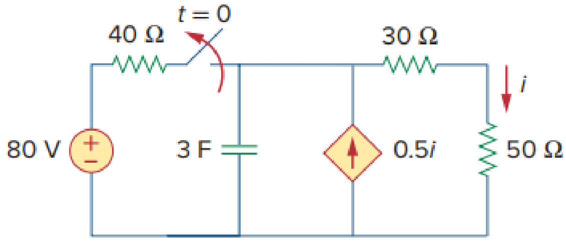
Chapter 7, Problem 43P

Problem

Consider the circuit in Fig. 7.110. Find $i(t)$ for $t<0$ and $t > 0$.

Fig. 7.110:



Step-by-step solution

Step 1 of 12

Refer to Figure 7.110 in the textbook.

At $t=0^-$, the circuit is at steady state. The capacitor acts as open circuit.

Draw the circuit diagram at $t=0^-$.

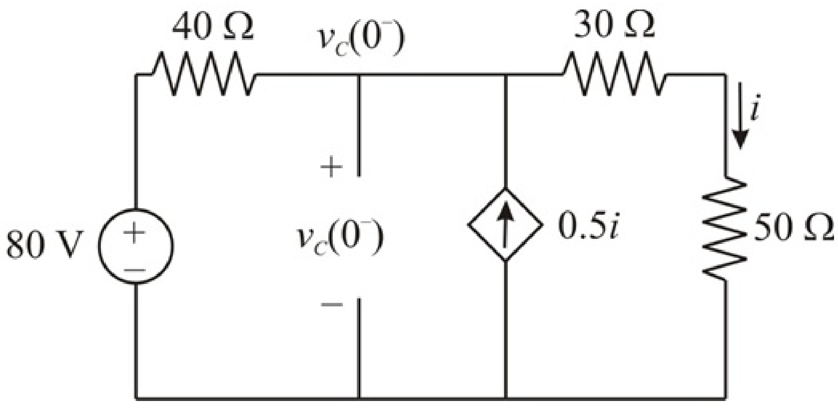


Figure 1

Step 2 of 12

Apply Ohm's law for the current through $50\ \Omega$ resistor.

$$i=\frac{v_C\left(0^-\right)}{30+50}$$
$$=\frac{v_C\left(0^-\right)}{80}$$

Apply Kirchhoff's current law at node $v_C\left(0^-\right)$.

$$\frac{v_C\left(0^-\right)-80}{40}-0.5i+i=0$$
$$\frac{v_C\left(0^-\right)-80}{40}+0.5i=0$$

Step 3 of 12

Substituting $\frac{v_C(0^-)}{80}$ for i.

$$\frac{v_C(0^-)-80}{40}+0.5\left[\frac{v_C(0^-)}{80}\right]=0$$

$$\frac{2\left[v_C(0^-)-80\right]+0.5v_C(0^-)}{80}=0$$

$$2v_C(0^-)-160+0.5v_C(0^-)=0$$

$$2.5v_C(0^-)=160$$

$$\begin{aligned}v_C(0^-)&=\frac{160}{2.5}\\&=64\text{ V}\end{aligned}$$

Step 4 of 12

The capacitor voltage cannot be changed instantaneously.

$v_C(0^-)=v_C(0^+)=v_C(0)=64\text{ V}$.

Calculate the current, i for $t<0$.

$$i=\frac{v_C(0^-)}{80}$$

Substituting 64 V for $v_C(0^-)$.

$$\begin{aligned}i&=\frac{64}{80}\\&=0.8\text{ A}\end{aligned}$$

Thus, the current, i for $t<0$ is $\boxed{0.8\text{A}}$.

Step 5 of 12

For $t>0$, the switch is open. Draw the modified circuit diagram.

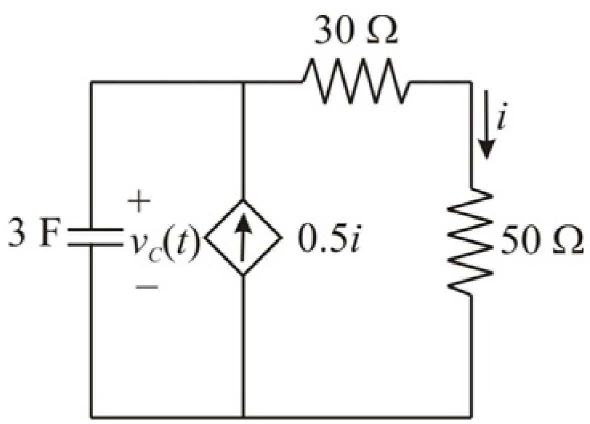


Figure 2

Step 6 of 12

At $t=\infty$, the circuit is at steady state and the capacitor behaves like an open circuit.

Draw the modified circuit diagram to calculate the final voltage across the capacitor.

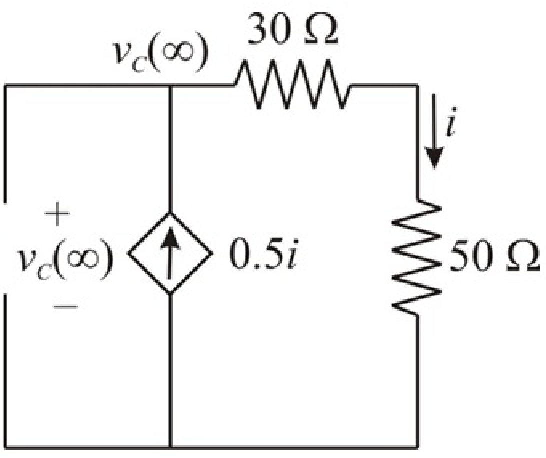


Figure 3

Step 7 of 12

The circuit is source free that is it does not have independent sources. So, the open circuited capacitor voltage is zero.

$$v_C(\infty)=0 \text{ V}$$

Consider the expression for the capacitor voltage.

$$v_C(t)=v_C(\infty)+\left[v_C(0)-v_C(\infty)\right]e^{-\frac{t}{\tau}}$$

Here, the time constant is $\tau=R_{\text{Th}}C$

Step 8 of 12

To calculate R_{Th}, remove the capacitor and connect test current source and determine equivalent resistance.

Draw the modified circuit diagram.

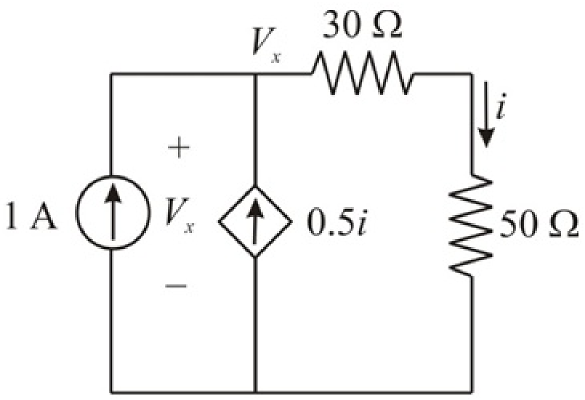


Figure 4

Step 9 of 12

Apply Ohm's law to calculate the expression for i.

$$i = \frac{V_x}{30+50} = \frac{V_x}{80}$$

Apply Kirchhoff's current law at node V_x.

$$-1 - 0.5i + i = 0$$
$$-1 + 0.5i = 0$$
$$0.5i = 1$$

Substitute $\frac{V_x}{80}$ for i.

$$0.5\left(\frac{V_x}{80}\right) = 1$$
$$V_x = 160$$

Step 10 of 12

Calculate the value of Thevenin's resistance, R_{Th}.

$$R_{\text{Th}} = \frac{V_x}{1\ \text{A}} = \frac{160}{1} = 160\ \Omega$$

Calculate the value of time constant.

$$\tau = R_{\text{Th}} C = (160)(3) = 480\ \text{s}$$

Calculate the expression for capacitor voltage.

$$v_C(t) = v_C(\infty) + \left[v_C(0) - v_C(\infty)\right] e^{-\frac{t}{\tau}} = 0 + (64 - 0) e^{-\frac{t}{480}} = 64 e^{-\frac{t}{480}}\ \text{V}$$

Step 11 of 12

Draw the circuit for $t > 0$.

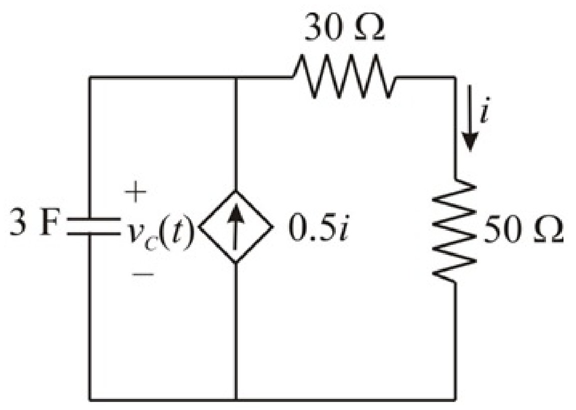


Figure 5

Step 12 of 12

Calculate the expression for the current, i.

$$i = \frac{v_C(t)}{30+50}$$
$$= \frac{v_C(t)}{80}$$

Substituting $64e^{-\frac{t}{480}}\ \mathrm{V}$ for $v_C(t)$.

$$i = \frac{64e^{-\frac{t}{480}}}{80}$$
$$= 0.8e^{-\frac{t}{480}}\ \mathrm{A}$$

Thus, the current, i for $t > 0$ is $\boxed{0.8e^{-\frac{t}{480}}\ \mathrm{A}}$.